



Trapping sound

Researchers have gained control of the "particle" of sound, the phonon. <https://digit.in/oct19-57>



Calcium-based batteries?

Calcium could replace lithium in batteries that store solar and wind power and could be cheaper too. <https://digit.in/oct19-58>

important for prey, even if they are immobile and hiding. A leopard for example, enjoys the advantage of stealth and vision at night. It is always better for an animal to be alert and aware of its surroundings. During the stages of deep sleep, sensory perception is cut off, making the animals more vulnerable to nocturnal predators. Sleep is observed even in animals where it is clearly counter productive, such as the marine mammals. This is one of the earliest theories of sleep, but it is unclear if a period of inactivity has any evolutionary advantage at all.

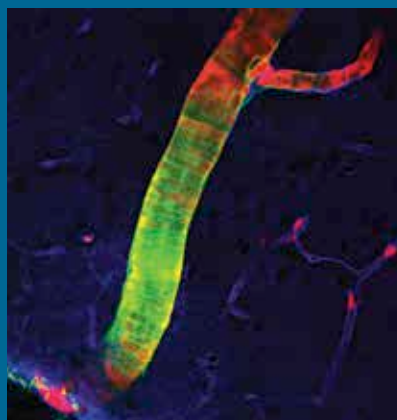
ENERGY CONSERVATION

Generally, it takes a lot more energy to be a carnivore as compared to a herbivore. A cheetah for example, has to expend more energy every time it feeds, as compared to a gazelle. According to the conservation theory, sleep is a mechanism that allows animals to reduce the amount of energy that they would otherwise consume if they remained awake. By going to sleep when not undertaking any essential tasks, such as foraging for food, sleep allows animals to use the energy that they have most efficiently. Competition for food is one of the most basic aspects of a struggle for existence, and sleep allows animals to require less food. We do know that when animals go to sleep, the temperature of the body falls, and the amount of energy consumed is also reduced. Sleep seemed to have emerged independently in endothermic animals that regulate their body temperatures. Birds for example, stand to gain considerably from energy conservation. There is also a positive correlation between sleep duration and metabolic rates... the animals that sleep the longest conserve the most energy. Extremely tiny animals such as shrews, squirrels and bats tend to lose a lot of heat very quickly because of their low body mass, and these are the species that tend to sleep longer in an effort to conserve energy. Infants of most animals also sleep much longer than the adults. The record holder is the Koala, which can sleep for up to 22 hours a day. However, these

animals could also maintain their body temperatures by remaining active and scouting for food. At the end of the day, this theory of sleep seems too much like a post hoc justification, instead of a complete explanation.

REST AND RESTORATION

This is something of a no-brainer, sleep allows us to rest, and that serves a function. It also explains why we wake up fresh and revitalised. However, this begs the question, what are the exact physiological processes that take place in the body during periods of sleep? According to the theory, sleep allows the body to undertake housekeeping activities, such as producing new cells, synthesizing proteins, growing muscles,



CSF flowing in a channel around the artery of a mouse brain

repairing wounds and flushing out toxic chemicals. One of the biggest findings to support this theory is how the immune system of all animals begin to deteriorate when deprived of sleep. The release of growth hormones for example, occurs mostly during periods of sleep.

Sleep also allows the brain to refresh. Throughout the day, the normal activity of neurons firing in the brain leads to a build up of a chemical known as adenosine. It is this accumulation that makes us feel tired, and even drowsy. Caffeine blocks the action of adenosine in the brain, which is why it is used to drive sleep away. During sleep, the adenosine is removed from the brain, which is why we wake up fresh. In a study looking at anaesthetised and sleeping mice,

researchers found an increase in flow of blood and lymph in the brain, a process they believe is related to the removal of waste of toxic proteins. The researchers used tracer molecules injected in the cerebrospinal fluid and found to their surprise that the fluid was moving through a brain in channels surrounding blood vessels. Scientists did not expect the brain to have such a plumbing system, and believed that the toxins were flushed out through a process of diffusion. The process occurs only in sleeping mice.

NEUROPLASTICITY

Brain plasticity involves understanding of how the structures of the brain change over lifetimes. This is somewhat of a mystery in its own right, as all the effects are not entirely known. However, observations of human infants have shown that their long duration of sleep is linked to the development of the brain. There is evidence of some links between sleep and brain plasticity in adults as well. It directly affects the ability of humans to learn new things and perform tasks. There are theories to suggest that sleep allows the brain to commit things to memory. This theory is extremely complex to investigate, because science does not have the tools or techniques yet to directly observe humans storing and retrieving memory, and it is not as simple as a read/write operation on hard disks. Broadly speaking, the processes are believed to involve acquisition, consolidation and recall. Of these three, sleep is believed to play a role in consolidation. There is some research that indicates that during sleep, the neurons grow and forge connections, solidifying memories and logic. Researchers are looking into brainwaves to check if there are certain stages of sleep that can be directly linked to forming some types of memories.

It is unlikely that there is only one function of sleep, and it is very possibly a combination of these theories or others. The topic is still an ongoing subject of research, and we continue to keep learning more about the function and role of sleep. 